

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q15-0533

Slocumb 100 kV  
January 03, 2018

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0533. The Interconnection Customer proposes a 126.3 MW solar photovoltaic facility interconnecting at the Slocumb 100 kV substation in the Ohio area. The request was received on November 01, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 178 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,292,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: DEKALB 230-1 kV Transformer Bank  
Contingency: DEKALB / RICHLAND 500 KV Line  
Loading: 133.5% (Rating: 1,082 MVA)  
DFAX: 7.5%  
Pre-project loading: 132.6%
2. Facility: BLUBONNET / RUSSELL FORK 230 KV line  
Contingency: JONS CHAPEL / DEKALB 500 KV  
Loading: 120.6% (Rating: 1,037 MVA)  
DFAX: 6.1%  
Pre-project loading: 119.9%
3. Facility: BLUEBONNET / JONES CHAPEL SUBSTATION 230 KV  
Contingency: RUSSELL FORK / ROBISON PARK SUBSTATION 230 KV L  
Loading: 109.8% (Rating: 1,085 MVA)  
DFAX: 5.0%  
Pre-project loading: 109.3%
4. Facility: JONES CHAPEL 500/230kV XFMR  
Contingency: JONES CHAPEL / DEKALB 500 KV  
Loading: 102.4% (Rating: 1,337 MVA)  
DFAX: 5.6%  
Pre-project loading: 101.9%

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

#### ***4. Required Network Upgrades and Cost Estimates***

| Facility                                    | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------|---------------------------------------------|---------------|-----------------------|
| DEKALB SUBSTATION 230 kV/1 kV Auto          | Replace transformer with higher-rated unit  | \$12,364,000  | 3% (\$393,000)        |
| BLUBONNET / RUSSELL FORK 230 KV line        | Rebuild line with higher-capacity conductor | \$60,935,000  | 3% (\$1,779,000)      |
| BLUEBONNET / JONES CHAPEL SUBSTATION 230 KV | Reconductor line                            | \$14,203,000  | 3% (\$482,000)        |
| JONES CHAPEL 500-230 kV Transformer         | Replace transformer with higher-rated unit  | \$22,080,000  | 3% (\$638,000)        |
|                                             | Total allocated to project                  |               | \$3,292,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.